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Research Article

Migration speed, timing, and long-term shifts in age structure in North American passerines during fall migration

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Migration is challenging for birds, especially juveniles, who experience high mortality rates during migration. The challenge is exacerbated in the Anthropocene, contributing to widespread population declines. Conservation efforts focused on increasing juvenile survival could bolster population recovery. Understanding how age structure of the migrant community shifts throughout migration could inform conservation efforts and future questions of migration ecology. However, it is unknown whether the age structure of the migrant community shifts spatially or temporally during migration. To answer these questions, we first analyzed age-related differences in migration speed and timing of departure during fall migration using 6 567 747 banding encounters, as variability in these components of migration could generate shifts in community demographics. We found widespread differences in migration speed (km d^{-1}) with adults being faster than juveniles in most species, and departure timing differences tied to adult molt. Our analyses revealed shifts in community demographics, with the proportion of juveniles within the community decreasing at northerly latitudes throughout migration. We also determined that demographics have shifted over 53 years, with the proportion of juveniles increasing in the north, and decreasing in the south. Our findings contribute to our knowledge of migration ecology, and our understanding of community shifts over time.

Keywords: age, avian conservation, banding data, demographics, differential migration, fall migration, migration speed, phenology

Introduction

Migration is challenging for migratory birds (Alerstam 2011, Klaassen et al. 2014, Rockwell et al. 2017). During migration, individuals face numerous threats and stressors, leading to mortality rates 4.4 times greater than during stationary periods (Sillett and Holmes 2002, Klaassen et al. 2014, Sergio et al. 2019, Newton 2025). Many of these threats have intensified with climate change and other human-mediated impacts

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in the Anthropocene, driving widespread declines in migratory bird populations (Sanderson et al. 2006, Bairlein 2016, Rosenberg et al. 2019, Leroy et al. 2026). Migration is particularly challenging for naïve juveniles undertaking their first migration (Sergio et al. 2019). Across all age classes, mortality risk is highest for juveniles during their first fall migration (Sergio et al. 2019, Newton 2025), and multiple studies have identified juvenile survival during this first migration as a critical driver of avian population dynamics (Clark and Martin 2007, Rushing et al. 2017, Wilson et al. 2018, Sergio et al. 2019, Seavy et al. 2025). Thus, juvenile survival during the first fall migration may represent a significant bottleneck in recovering declining migratory avifauna (Wilcove and Wikelski 2008, Klaassen et al. 2014). Conservation efforts aimed at increasing survival during this perilous life stage, such as stopover habitat restoration (Cohen et al. 2021, Linscott and Senner 2021, Schmaljohann et al. 2022), or adaptive management of infrastructure such as 'lights out' campaigns (Van Doren et al. 2017, 2021, Burt et al. 2023), and wind energy curtailment (Marques et al. 2014, Cohen et al. 2022, Bradarić et al. 2024), may play a crucial role in recovering populations and the ecosystem services they provide. Given the immense scale of migration, and limited resources for conservation, efforts must be strategically targeted to maximize impact (Joseph et al. 2009, Wilson and Law 2016, Walls 2018, Horton et al. 2021, Seavy et al. 2025). Identifying where and when the proportion of juveniles within the migrant community peaks during migration and then focusing targeted conservation efforts on these time periods and locations may provide a disproportionate benefit to population recovery (Sæther and Bakke 2000, Clark and Martin 2007, McKim-Louder et al. 2013, Oppel et al. 2015, Rushing et al. 2017, Wilson et al. 2018, Sergio et al. 2019, Seavy et al. 2025). However, it is currently unknown whether the age structure of the migrant community shifts spatially or temporally during fall migration.

Age-related differences in migration speed, timing of departure from the breeding grounds, routes, and non-breeding ranges, along with higher juvenile mortality, could collectively contribute to demographic variability in the migrant community throughout the course of a migration season. For instance, juveniles often spend the non-breeding season at different latitudes than adults (Cristol et al. 1999, Woodworth et al. 2016), or are more likely to stopover closer to coastlines (Ralph 1981, Morris et al. 1996, Woodrey and Moore 1997, Payevsky 1998), which may lead to shifting demographics across space. Age-related differences in migration speed (Ellegren 1993, Fransson 1995, Mitchell et al. 2015, Boyd et al. 2025), and phenology (Woodrey and Chandler 1997, Stewart et al. 2002, Carlisle et al. 2005, Morris et al. 2016, Covino et al. 2020, Wobker et al. 2021, Neate-Clegg and Tingley 2023, Brempt et al. 2025), have been documented across a range of species and systems, providing important insights into age-specific migration strategies. Much of this previous work, particularly for migration speed, has generally focused on individual species, or limited geographic regions, limiting our community-level understanding of these dimensions of migration (however, see Neate-Clegg and Tingley (2023) for a broad-scale analysis of phenological differences across species). Consequently, whether age-related differences in migration speed or phenology are sufficiently widespread across species to influence demographic patterns at the community level remains unclear. An improved understanding of differences in migration strategy between age classes, particularly as it relates to migration speed, would inform our understanding of how differences in these strategies may lead to community-level demographic variability in migration.

Here we leveraged 6 567 747 unique bird banding encounters during fall migration from across North America over 53 years to test for age-related differences in migration speed and timing of departure from the breeding grounds (hereafter, timing) for 69 species of passerine migrants and assessed whether age structure of the migrant community shifts across space and time during fall migration and across years. For migration speed and timing, we hypothesized that speed is age dependent and predicted that adults travel faster than juveniles (Ellegren 1993, McKinnon et al. 2014, Mitchell et al. 2015), and that timing varies with age and is linked to adult flight feather molt strategy (Woodrey and Chandler 1997, Carlisle et al. 2005, Stutchbury et al. 2010, Neate-Clegg and Tingley 2023). We predicted that median passage date would be later for adults relative to juveniles in species where adults molt flight feathers on the breeding grounds, and earlier in species where adult flight feather molt occurs on non-breeding grounds or during migration (Woodrey and Chandler 1997, Butler et al. 2002, Carlisle et al. 2005, Kiat and Izhaki 2016, Neate-Clegg and Tingley 2023). For our analysis of migrant community demographics, we hypothesized that age-mediated differences in migration speed and timing across species would result in shifting age structure of the migrant community throughout fall migration and across space. Specifically, we predicted that the proportion of juveniles in the community would decline at northerly latitudes and increase at southerly latitudes as migration progresses, since adults typically overwinter further to the north. Lastly, we hypothesized that climatic trends have created shifts over time in migrant demographics at a continental scale, possibly through changes in non-breeding distributions or variation in juvenile mortality over time (Cristol et al. 1999, La Sorte and Thompson III 2007, Lehikoinen et al. 2013, Gillings et al. 2015, Woodworth et al. 2016). From these analyses, we identify periods and locations where the proportion of juveniles in the community is high, which could be targeted to maximize the impact of conservation strategies aimed at increasing juvenile survival.

Material and methods

Bird banding data handling

United States Geological Survey Bird Banding Lab (BBL) data from 1970 to 2023 were obtained for common North American passerine migrants via the 2023 release, which

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included data through spring 2023 (Nakash et al. 2023). We downloaded encounter data (both banding and recapture) for all species from the families Tyrannidae, Vireonidae, Turdidae, Mimidae, Passerellidae, Parulidae, and Cardinalidae. These families were chosen as they comprise the majority of species and individuals encountered in typical banding efforts. We filtered the raw BBL data following methods employed by previous studies using BBL data to study migration phenology (Covino et al. 2020, Neate-Clegg and Tingley 2023). We excluded species with limited ranges in the US and Canada, or with highly overlapping breeding and non-breeding ranges. For all remaining species we removed encounters outside of the sampling area (25–70°N and 160−50°W), or outside of the fall migration period (10 August–15 November). We retained only encounters with a 'bird status' of 3 (normal wild bird) and removed encounters where 'age code' was 0 (unknown), 3 (juvenile, used for nestlings and recent fledglings), or 4 (local). Retained encounters were labelled as juvenile (age code 2, corresponding to individuals in their hatch year undergoing their first fall migration), or adult (age codes 1, 5, 6, 7, or 8). We retained only the first encounter for birds captured multiple times at the same location during the same fall migration season.

We created 0.5 × 0.5° sampling bins for the entirety of the sampling area and associated each encounter with the relevant latitude/longitude sampling bin. Then, for each species, we only retained bins where the species was encountered on 25 unique ordinal days and a minimum of 10 individuals of each age class were captured. Finally, we removed species that were encountered in fewer than 5 of the 0.5° latitudinal bands and within fewer than 15 sampling bins. Summary data of distribution of sampling locations, the number of encounters for each species, number of birds of each age class, and number of unique sampling bins are provided in the Supporting Information. In total, 6 567 747 banding encounters from 69 species and 672 sampling bins were retained for analyses (Supporting information).

Analysis of migration speed and timing

For each 0.5 × 0.5° sampling bin, for each species, we calculated the ordinal date at which median passage occurred (median passage date) for the fall migration season with all years of BBL data combined (Fig. 1a). Median passage date was defined as the ordinal date when half of the cumulative sum of migrants of a given species had been encountered within the sampling bin (Covino et al. 2020). For each species, we then fitted a linear model with median passage date (unique to each sampling bin) as the response variable, and interaction terms between latitude (corresponding to the southern edge of the sampling bin) and age class (either juvenile or adult, modeled as a factor), and longitude (corresponding to the western edge of the sampling bin) and age class as predictors. Latitude and longitude were scaled and centered prior to modeling. To account for differences in sampling effort and sample size across sampling bins and age classes, we weighted each observation in the model based on the estimated uncertainty of its median passage date, such that observations with greater sample sizes and more precisely estimated median passage dates contributed proportionally more to the model. To calculate these weights, we estimated the probability density function of the distribution of passage dates for each observation in the model using a Gaussian kernel density estimator and obtained the value of the density at the median passage date $\left(f\left(\tilde{x}\right)\right)$. We then used this value to calculate the estimated standard error of the median passage date using the equation:

$$SEmed = 1 / \left(2f\left(\tilde{x}\right)\sqrt{n}\right)$$

where n is the total number of individuals of that age class and species captured in the sampling bin. Model weights (w) were then calculated as the inverse of the variance of the median passage date using the equation:

$$w = 1 / SEmed^2$$

Following this initial modeling step, we assessed whether species should be retained for further analyses based on the modeled relationship between median passage date and latitude. A lack of relationship between these two variables could indicate widespread variability in migration timing, or limitations of the dataset causing a true relationship to be undetectable, meaning that the speed and timing estimates derived from these models would be unreliable. Species were removed if the adjusted p-value (calculated with the Benjamini–Hochberg method to control for multiple comparisons) was > 0.05, indicating that latitude is not a significant predictor of median passage date.

We used the output from these models to compare migration speed and timing between age classes for each species. Specifically, for each species, we calculated migration speed for each age class based on the relationship between latitude and median passage date as estimated by the model. For adults, the coefficient estimate of the scaled latitude term (β_{lat}) represents the change in median passage date for a one standard-deviation increase in latitude. The following equation was used to derive migration speed (S_{AHY}; km day^{-1}) for adults by transforming the latitude variable back into the original scale using the standard deviation of the original unscaled latitude variable (σ_{lat}), converting from degrees latitude to kilometers (111 km degree^{-1}):

$$S_{AHY} = \frac{111}{\left|\left(\beta_{lat}\right) / \sigma_{lat}\right|}$$

For juveniles, the change in median passage date for a one standard-deviation increase in latitude is equal to the sum of the coefficient estimate for the scaled latitude term (β_{lat}) and

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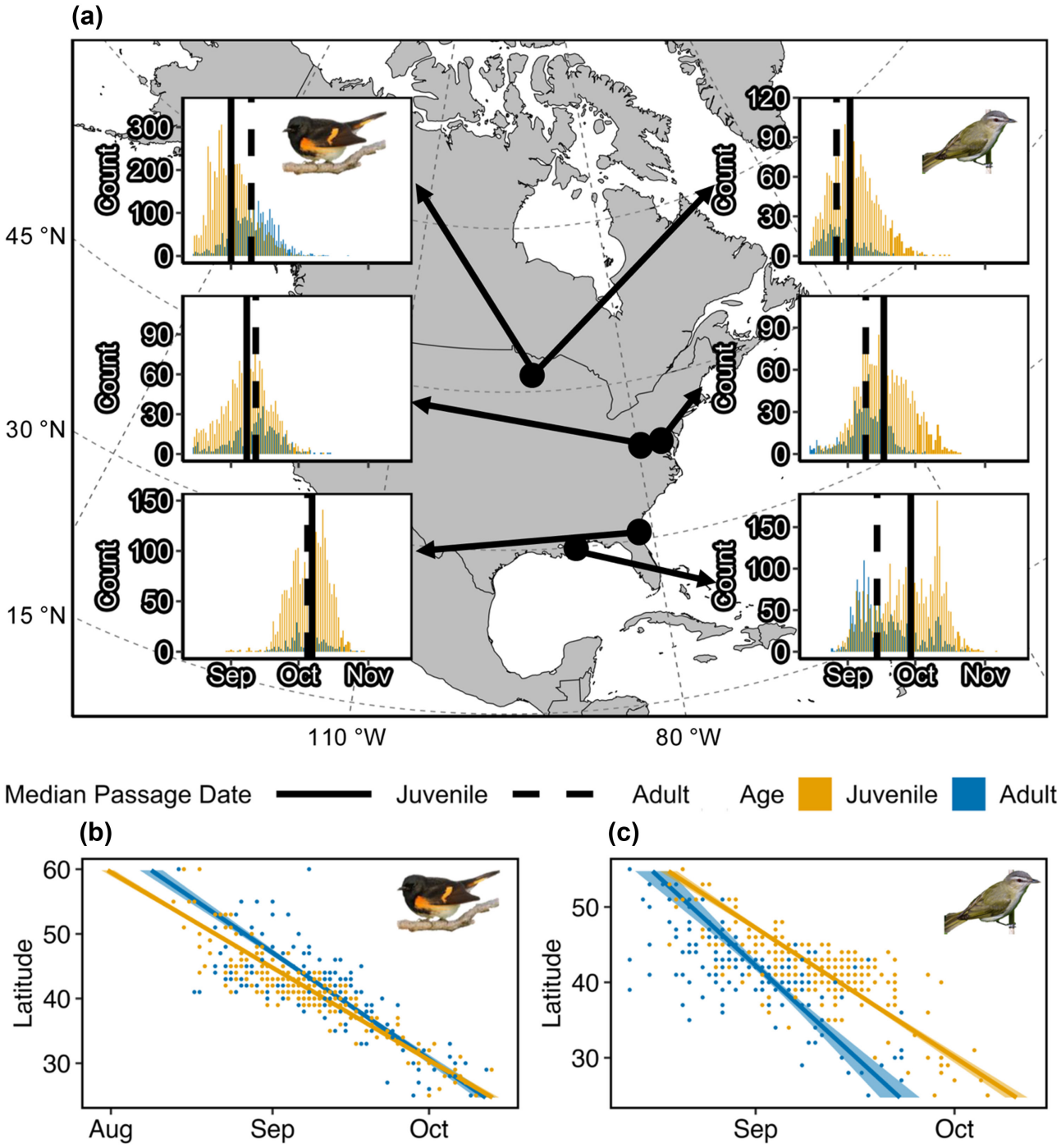


Figure 1. A representation of the speed and timing of migration analysis for juvenile and adult age classes of *Setophaga ruticilla* (American redstart) and *Vireo olivaceus* (red-eyed vireo) where (a) depicts fall migration passage and median passage date for each age class for each species (*S. ruticilla* on left, *V. olivaceus* on right) for a set of locations across a range of latitudes, (b) and (c) depict the modeled median passage date as a function of age class and latitude for each species with 85% confidence intervals and dots representing median passage date and latitude for each sampling bin. The same analyses were completed for each of the 57 retained species included in this study, and model output for each species can be found in the Supporting information. Bird photos from Macaulay Library at Cornell Lab of Ornithology (*S. ruticilla*, ML28948631; *V. olivaceus*, ML161670801).

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the latitude by age class interaction term ($\beta_{lat:Age}$). The following equation was used to derive migration speed (S_{HY}; km day^{-1}) for juveniles:

$$S_{HY} = \frac{111}{\left|\left(\beta_{lat} + \beta_{lat:Age}\right) / \sigma_{lat}\right|}$$

Standard errors for migration speed were calculated using the variance-covariance matrix of the model coefficients, applying the delta method to account for uncertainty in both the main effect and the interaction term. The difference in migration speed between age classes was calculated by subtracting S_{HY} from S_{AHY}.

To estimate timing differences between age classes we used the model to predict median passage date for each age class at the third quartile of latitude. We chose to use the third quartile of latitude, rather than the mean or median, as timing differences estimated for latitudes closer to the breeding grounds should be less sensitive to migration speed differences between age classes (Fig. 1b, c). Timing difference was calculated by subtracting the median passage date for juveniles from the median passage date for adults. To assess the ability of our approach to estimate migration speed, we compared migration speed as estimated herein to migration speed as measured via geolocator- or MOTUS-based methods for species where possible.

We assessed whether variation in adult flight feather molt strategy explained variation in migration timing using a phylogenetic generalized least-squares model. The coefficient estimate for the age class term for each of the 57 species-specific speed/timing models was the response variable and adult molt strategy was the predictor. We placed migrants into the categories of non-breeding (adult flight feather molt occurs on non-breeding grounds), fall (species with molt-migration or flight feather molt occurring on both breeding and non-breeding grounds), or breeding (adult flight feather molt occurs on breeding grounds) based on their adult flight feather molt strategy (Pyle et al. 2018, Neate-Clegg and Tingley 2023). We fitted the model using the *gls()* function from the R package 'nlme' and obtained estimated marginal means for each molt strategy and conducted Tukey-adjusted pariwise comparisons using the *emmeans()* function from the R package 'emmeans' (Pinheiro et al. 1999, Lenth et al. 2025). We controlled for phylogenetic relationships in this model by defining a phylogenetic correlation structure. To do this, we used the *get_trees()* function from the R package 'rtrees' to download phylogenetic trees limited to the species included within the model (Li 2023). The *get_trees()* function returns a subset of 100 posterior phylogenies, the first 50 of which are randomly selected from the 10 000 trees with Ericson backbone generated by Jetz et al. (2012), and the second 50 are randomly selected from the 10 000 trees with Hackett backbone. We then created a consensus tree using the *ls.consensus()* function from the R package 'phytools' with a majority-rule threshold of 0.5 (Revell 2024). We then used the consensus tree to define a Brownian phylogenetic correlation structure for the model using the *corBrownian()* function from the R package 'ape' (www.r-project.org, Paradis et al. 2004).

Analysis of migrant community age structure

We modeled the proportion of juveniles using a generalized additive mixed model with a quasi-binomial family implemented using the *bam()* function from the R package 'mgcv' (www.r-project.org, Pedersen et al. 2019). Using the processed BBL data, which included all 69 species, for each sampling bin and year, we calculated the number of juveniles captured and the number of adults captured for each week of the fall migration season with all species combined. The response variable for the model was specified as a two-column matrix of counts (juveniles and adults), such that the model treated the proportion of juveniles as the probability of 'success' in a binomial trial. We included the predictor variables of week, year, latitude, and longitude, as well as interactions between week and latitude, week and longitude, year and latitude, and year and longitude. Week and year were included as smooth terms to capture within-year and across-year temporal trends, respectively, with smoothing parameters (k) set to 5. The interaction terms were modeled using tensor product smooths (k = 3) to account for nonlinear, spatiotemporal interactions. To account for spatial autocorrelation, we included a bivariate Gaussian process smooth over latitude and longitude. To account for temporal autocorrelation, we specified AR.start in the model which indicates the start of each time series within a bin and allows the model to adjust for non-independence between successive weeks. Lastly, we included a random intercept and slope for week, grouped by sampling bin, to account for variation in baseline proportion juvenile across bins and for differences in temporal changes among bins. Using this model, we then calculated the seasonal change in the proportion of juveniles within the migrant community by subtracting the predicted proportion of juveniles for the first week of the migration season from the predicted proportion of juveniles for the final week of the migration season, for all 0.5 × 0.5° sampling bins located within 4° of latitude or longitude of a sampled bin. From these data, we created a map depicting seasonal change in proportion juvenile across the study area. All analyses were conducted and figures generated using R ver. 4.4.3 (www.r-project.org).

Results

Age-based differences in migration speed and timing

Following the initial modeling step, 12 species were removed from the dataset, and 57 species were retained (Supporting information). Migration speed was significantly different between juveniles and adults for 24 of the 57 species analyzed, with adults being faster in 19 species and juveniles being faster in five species (Fig. 2a, Supporting information). At least one species from each of the seven families analyzed had a significant difference in migration speed between age classes (Fig. 2a, Supporting information). Migration timing

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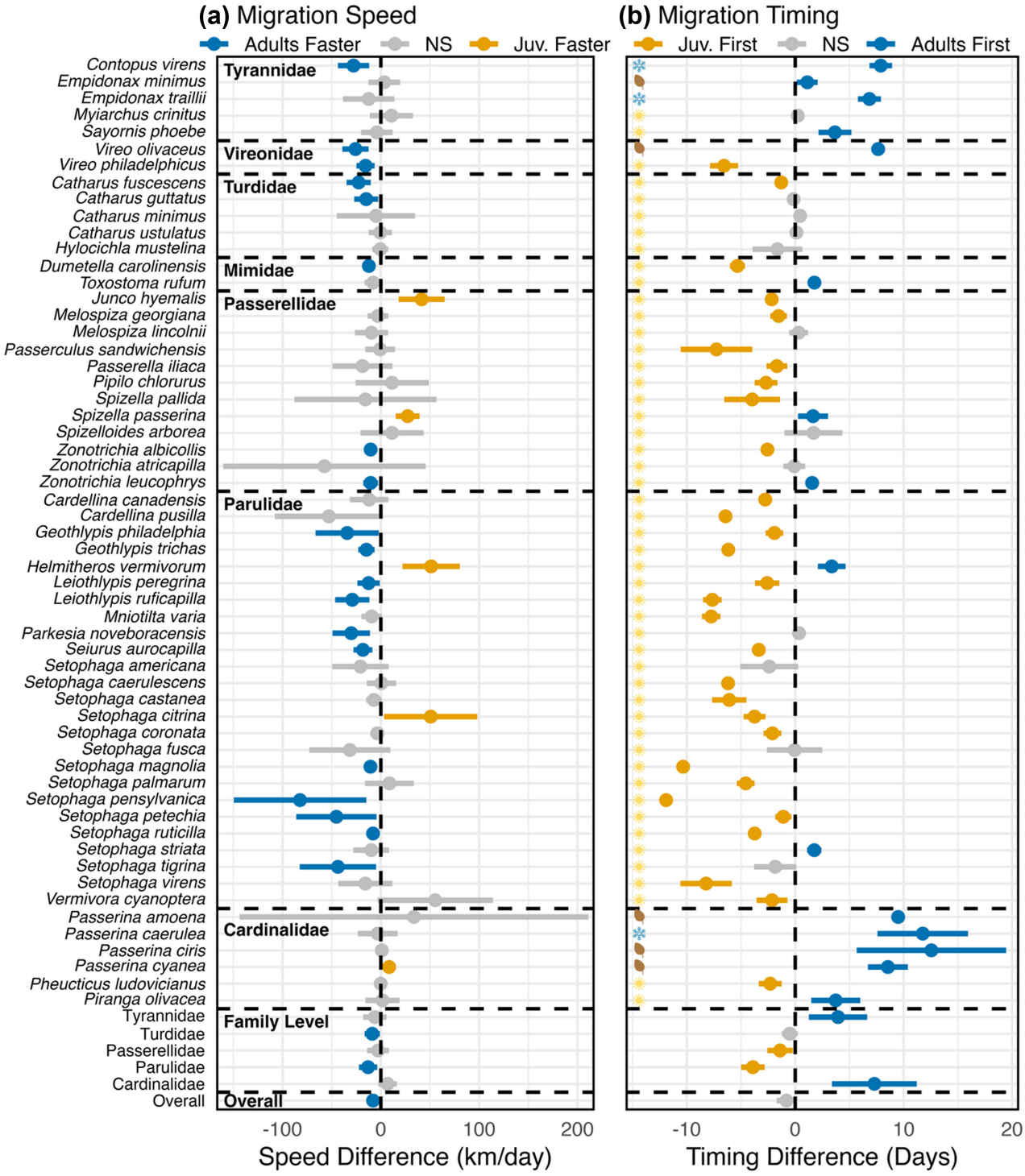


Figure 2. Differences in (a) migration speed (juvenile minus adult), and (b) migration timing (juvenile minus adult) between age classes for 57 species of North American passerine migrants from seven families. Adult flight feather molt strategy categories are represented by icons associated with each species with fall strategy represented by a leaf, non-breeding represented by a snowflake, and breeding represented by a sun. Coefficient estimates are presented with 85% confidence intervals for each species. Averaged coefficient estimates and 85% confidence intervals are provided for each family that included more than two species, and overall. Full model output for each species can be found in the Supporting information.

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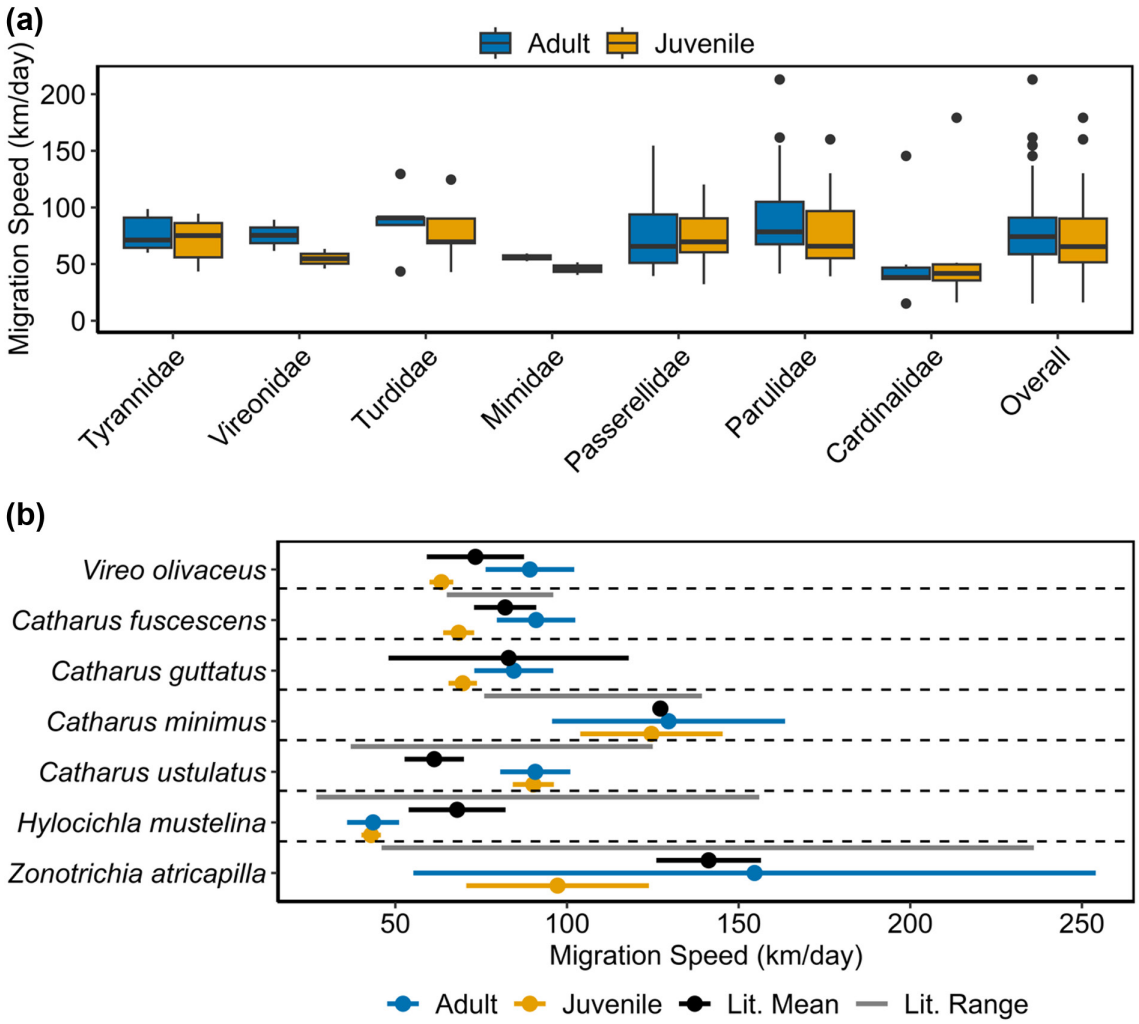


Figure 3. Comparison of migration speed between (a) each of the families studied and overall, and (b) between estimates derived from bird banding data from our study, and from geolocator and MOTUS based estimates from the literature. In (b) migration speed estimates for adults and juveniles are included with 85% confidence intervals (in blue and orange, respectively), mean values from the literature are included with relevant uncertainty measures (in black; see the Supporting information), as well as the range of values from the literature (in gray). References for data sources are included in the Supporting information.

was significantly different between juveniles and adults for 45 of the 57 species analyzed, with juveniles migrating first in 30 species and adults migrating first in 15 species (Fig. 2b, Supporting information).

Across all 57 species, average migration speed (± SD) was estimated to be 72.34 ± 31.44 km day^{-1} for juveniles and 80.26 ± 36.94 km day^{-1} for adults (Fig. 3a, Supporting information). Migration speed estimated herein closely mirrored migration speed estimates for species tracked with geolocators or MOTUS (Fig. 3b, Supporting information). Variation in adult flight feather molt strategy had a significant effect on differential migration timing by age classes, such that in species where adults molt flight feathers on the breeding grounds, juveniles migrated earlier than adults (Fig. 4). The timing difference was reversed for species with adult flight feather molt occurring on the non-breeding grounds and for species molting during migration; however, the difference between species molting on wintering grounds or during migration was not significant (Fig. 4).

Migrant community demographic shifts

The model assessing temporal and spatial variability in migrant community demographics accounted for 25.3% of the variation in the proportion of birds in the migrant community that are juveniles (Supporting information). All terms included in the model were identified as significant predictors of the proportion of juveniles within the migrant community (Supporting information). Throughout the migration season, and across years, the proportion of juveniles is higher at northern latitudes and eastern longitudes than southern latitudes and western longitudes (Fig. 5). The proportion of juveniles decreases at northerly latitudes throughout the migration season but is primarily stable at southerly latitudes, and trends are consistent across longitudes throughout the

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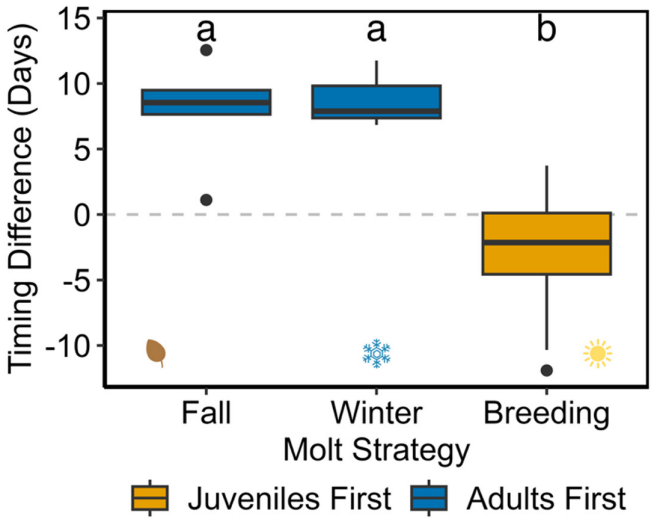


Figure 4. Boxplots depicting the difference in median passage date (days) between adults and juveniles for each of the three (fall, winter, breeding) adult flight feather molt strategies. Groups labelled with the same letter above the boxes do not differ significantly in migration timing, while groups labelled with different letters differ significantly.

season (Fig. 5a, b, 6). Across years, the proportion of juveniles has increased at northern latitudes and decreased in the south (Fig. 5c). Across 53 years, the proportion of juveniles has remained stable in the east and increased in the west (Fig. 5d). Using the model, we generated a map of seasonal change of the proportion of the migrant community that are juveniles by calculating proportion juvenile for every sampling bin for the first week of migration and subtracting this value from the predicted value for the final week of migration (Fig. 6).

Discussion

In this study, we used a large publicly available dataset of bird banding records to assess age-related differences in migration speed and timing of departure from the breeding grounds for 57 species of North American passerines and conducted a novel analysis of migrant community demographics throughout fall migration. We found that adults typically migrate faster than juveniles, and migration timing between age classes is explained in part by adult flight feather molt strategy. Also, we determined that the age structure of the migrant community shifts at a landscape scale throughout fall migration and, additionally, has changed over the last 53 years. Our findings provide insight into migration behavior,

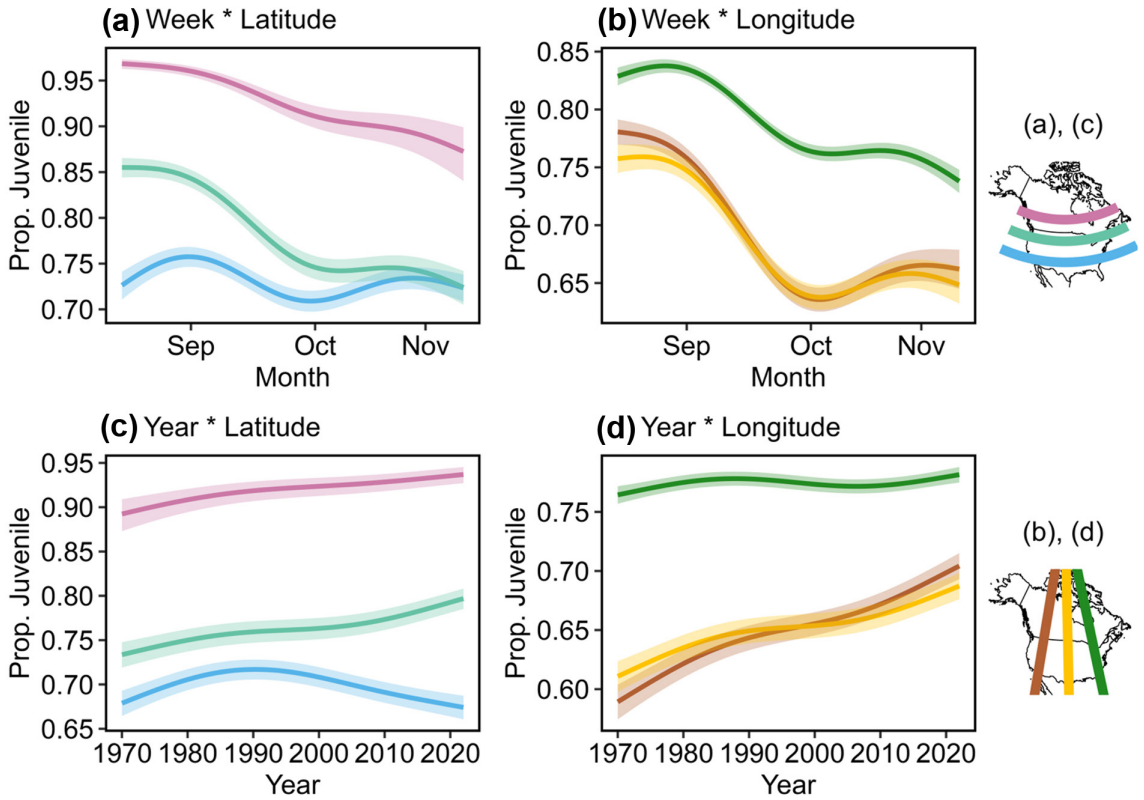


Figure 5. Community-level shifts in migrant age structure and the relationship between proportion juvenile and the interactions between (a) week and latitude, (b) week and longitude, (c) year and latitude, and (d) year and longitude. Trendlines were calculated for each predictor averaged over the full observed distribution of the other covariates in the model. All trend lines include 85% confidence intervals. Full model output can be found in the Supporting information.

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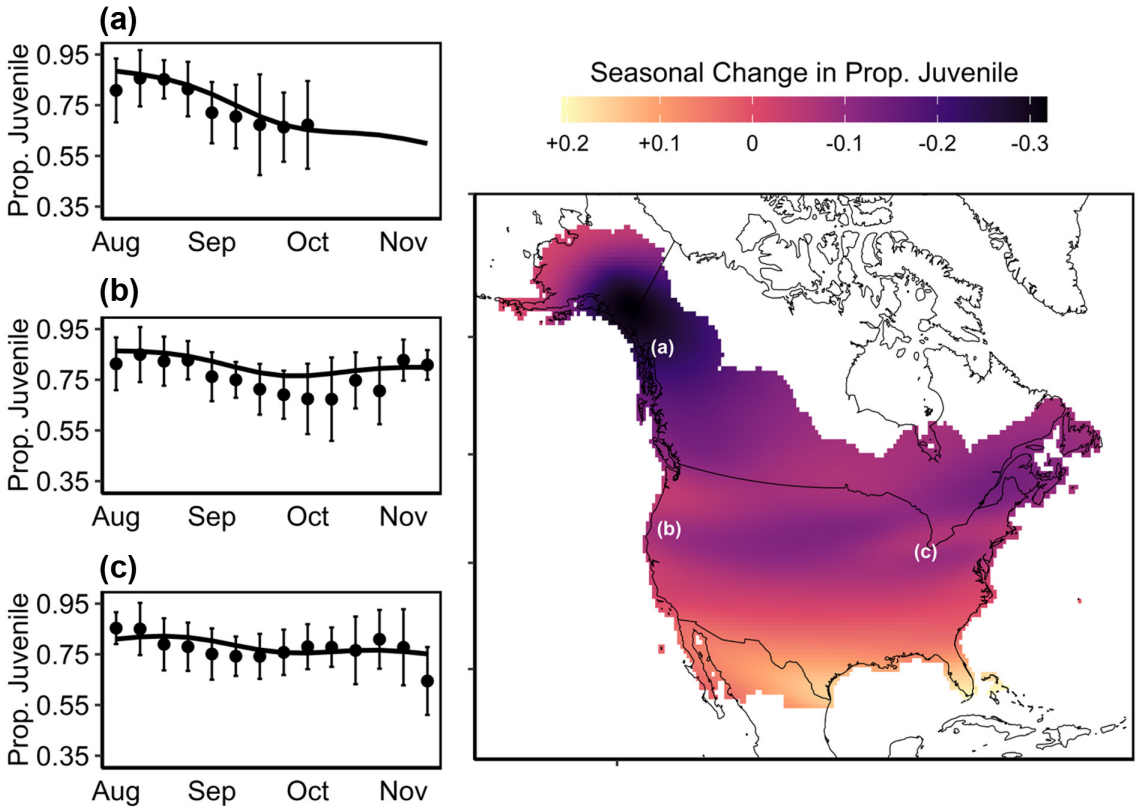


Figure 6. Map demonstrating the shifting demographics of the community of 69 species of North American passerine migrants throughout a fall migration season. Darker colors represent locations with stronger decreases in proportion juvenile throughout fall migration while lighter colors represent locations with no change or increasing proportion juvenile throughout fall migration. Insets are included for a set of sampling bins and depict points for the mean proportion juvenile over the 53 years of bird banding data for that specific week in that specific sampling bin, and error bars represent ± 1 SD from the mean, with trendlines depicting the modeled relationship between proportion juvenile and week of the migration season.

ecology, and determinants of phenology in North American passerines and elucidate previously undocumented shifts in the demographic structure of the avian migrant community throughout the fall migration season and across decades.

While bird banding data have been used previously to assess differences in migration timing between age classes, most studies have analyzed only a few species from a small number of locations (Woodrey and Chandler 1997, Carlisle et al. 2005, Morris et al. 2016, Woodworth et al. 2016, Covino et al. 2020, Wobker et al. 2021). Here, we were able to use banding data to analyze migration timing for many species at a continental scale and determine how timing differs between age classes. It is important to note, however, that our estimates of migration timing differences are likely unable to be fully disentangled from migration speed differences between age classes, and the latitude at which timing differences were calculated. For instance, for a given species, if both age classes leave the breeding grounds on the same date, but adults travel at faster speeds than juveniles, then our model would likely estimate that adults precede juveniles in migration since the timing difference is measured at a latitude to the south of the breeding grounds. Nonetheless, our analyses were successful in recovering estimates of timing that mirrored those of previous studies for individual species (Carlisle et al. 2005, Covino et al. 2020, Brempt et al. 2025), providing evidence that our analytical methods are capable of disentangling speed and timing in a meaningful way. As hypothesized, we found timing differences to be closely tied to adult flight feather molt strategy. Adults precede juveniles in migration for species where adults molt flight feathers during fall migration or on the wintering grounds, and this trend is reversed for species where adults molt flight feathers on the breeding grounds. Our findings further solidify our understanding of the close relationship between adult flight feather molt strategy and age-related differences in departure from the breeding grounds (Woodrey and Chandler 1997, Carlisle et al. 2005, Pyle et al. 2018, Neate-Clegg and Tingley 2023, Brempt et al. 2025).

To date, few studies have used bird banding data to estimate migration speed (Covino et al. 2020). Using bird banding data, we determined that differences in migration speed are widespread taxonomically, and that in nearly all cases where the difference was significant, adults migrate faster than juveniles, as hypothesized by previous authors (Newton 2011, 2023, Nilsson et al. 2013, Åkesson and Helm 2020). The congruence of our findings for individual species with

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those of previous studies (Fig. 3b) provides validation of our methods for analysis of speed of migration across species. Our estimates of migration speed using a population level approach through bird banding data are in much closer alignment to individually based measures than other population level approaches, such as those using eBird data (La Sorte et al. 2013). Researchers to date have relied primarily on telemetry, satellite tracking, or citizen science data to assess migration speed (La Sorte et al. 2013, Nilsson et al. 2013, Boyd et al. 2025), methods that are limited by a small sample size and/or the body size of the study species (Scarpignato et al. 2023). Our findings indicate that bird banding data may be useful in measuring migration speed at the species level, without the restriction of species needing to be of a certain size to be studied. However, for bird banding data, sample size is still a limiting factor for some species, and banding locations are not widespread enough to fully capture the entire annual distribution of each species. Therefore, as geolocator, satellite, and telemetry-based methods for the quantification of individual migration speed and timing become more widely accessible, it is likely that these approaches will serve greater utility than bird banding data.

We have demonstrated the novel findings that demographics of the avian migrant community shift both at a landscape scale over the course of a fall migration season and over the last five decades. These findings open new avenues for future inquiry regarding the behavior and ecology of avian migrants. For instance, a growing body of research is demonstrating the importance of social information provided by adults to juveniles during their first migration (Mueller et al. 2013, Aikens et al. 2024, Osterhaus et al. 2025a, Van Doren et al. 2025). Our findings indicate that the number of adults available from whom juveniles may obtain information changes throughout each migration season, and has changed over the past 53 years, with unknown ramifications.

Multiple studies have demonstrated generally northward changes (with some longitudinal variability) in non-breeding distributions of avian migrants in response to changing climate, but have not assessed differences in these shifts among age classes (La Sorte and Thompson III 2007, Lehikoinen et al. 2013, Gillings et al. 2015). The trends found here of latitudinal and longitudinal shifts in age structure over decades could be explained by greater northward and westward shifts in the non-breeding distribution of juveniles relative to adults, a hypothesis that should be pursued by future studies. Similarly, our findings could reflect shifting survival between age classes. For instance, across years, an increase in the proportion of juveniles in the north could be the result of either increased juvenile survival or reduced adult survival. It is also important to note that age structure information derived from bird banding data may not directly reflect the true age structure of migrant community populations, which could bias the results presented here (Ralph 1981, Dunn et al. 2004, Mills 2016). Specifically, in fall migration, it appears that capture data are biased towards juvenile birds (Mills 2016), and this may vary between habitats (Ralph 1981, Dunn et al. 2004), so the proportions presented herein should be considered with this fact in mind. It is possible that the trends in demographic structure detailed here could be the result of variability in capture probability between age classes driven by unknown factors. To account for these biases and improve the utility of demographic data derived from bird banding, future work must assess variability in capture probability between age classes (and across sexes, body condition, etc.), and how this varies through space and time.

Given the high rates of mortality experienced by naïve juveniles during their first migration, conservation efforts focused on improving juvenile survival during this critical life period may aid in the restoration of declining populations of avian migrants. Our findings demonstrate that there may be regions and time periods towards which efforts could be targeted to maximize impacts for juveniles. Pairing our findings with those of previous studies identifying stopover habitats with high densities of avian migrants (Guo et al. 2023, 2024, Horton et al. 2023) will be a useful strategy for identifying where and when conservation efforts should occur. By pairing these data, we can identify locations where juvenile abundance is high throughout the migration season and recommend these locations as priorities for conservation actions. Alternatively, locations with low proportions of juveniles in the population may represent habitats that are not able to support juveniles, and efforts could be focused to these areas to increase habitat quality for juveniles. However, current stopover maps are created at a seasonal level (Guo et al. 2023, 2024, Horton et al. 2023), while demographic data generated by our study are at the weekly level. Therefore, future work should aim to generate stopover abundance maps at the weekly level to improve our ability to target efforts towards areas and times that will bring maximum benefit for juvenile migrants.

The demographic shifts of the avian migrant community identified by our study may help explain seasonal trends in migrant behavior observed in other studies. For example, a study in New Mexico, USA, found a strong behavioral response to artificial light early in migration (August–September), but not later (October–November) (Osterhaus et al. 2025b). The authors proposed two explanations for this trend: 1) juveniles may respond more strongly to light than adults and may be more abundant in the community early in migration, leading to a higher early-season response, or 2) juvenile responsiveness declines as they develop. Our findings show a slight increase in the proportion of juveniles at this location throughout migration, which provides evidence against the first hypothesis and indicates that the second hypothesis should receive further study.

Migratory birds in the Anthropocene have experienced large-scale population declines. To reverse the decline of avian migrant populations, conservation efforts must focus on increasing survival and productivity throughout the annual cycle. However, even though specific conservation challenges have been identified (Seavy et al. 2025), our abilities to implement conservation efforts at scales matching that of avian migration are limited by time, funding, and

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workforce (Joseph et al. 2009, Wilson and Law 2016, Walls 2018). As such, identifying locations and time periods to prioritize for conservation action will be critical to maximize our ability to conserve avian migrants (Horton et al. 2021). Our findings demonstrate previously undocumented shifts in the demographic make-up of the avian migrant community at a landscape scale and, when paired with abundance data and maps of important stopover areas (Horton et al. 2021, 2023, Guo et al. 2023, 2024), may aid in prioritizing locations for conservation efforts targeted to increase juvenile survival. Additionally, as we progress towards a more holistic understanding of avian migration ecology and behavior at landscape scales with species-level resolution (Bowlin et al. 2010, Robinson et al. 2010, Van Doren and Horton 2018, Dokter et al. 2018, Gayk and Mennill 2020, Van Doren et al. 2023, 2025, Osterhaus et al. 2025a), our analysis of migration speed and timing between age classes, and the demographics of the migrant community throughout an entire migration season, add another key component to our understanding of migration.

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Author contributions

Dylan M. Osterhaus: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Funding acquisition (supporting); Investigation (lead); Methodology (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (equal). **Martha J. Desmond**: Conceptualization (supporting); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review and editing (equal). **Timothy F. Wright**: Conceptualization (supporting); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review and editing (equal).

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Data availability statement

Data are available from the Dryad Digital Repository: https://doi.org/10.5061/dryad.xd2547dvt (Osterhaus et al. 2026).

Supporting information

The Supporting information associated with this article is available with the online version.

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